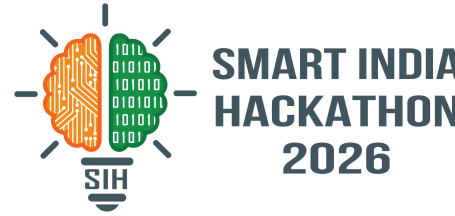


SMART INDIA HACKATHON 2026



Problem Statement ID - SIH26106

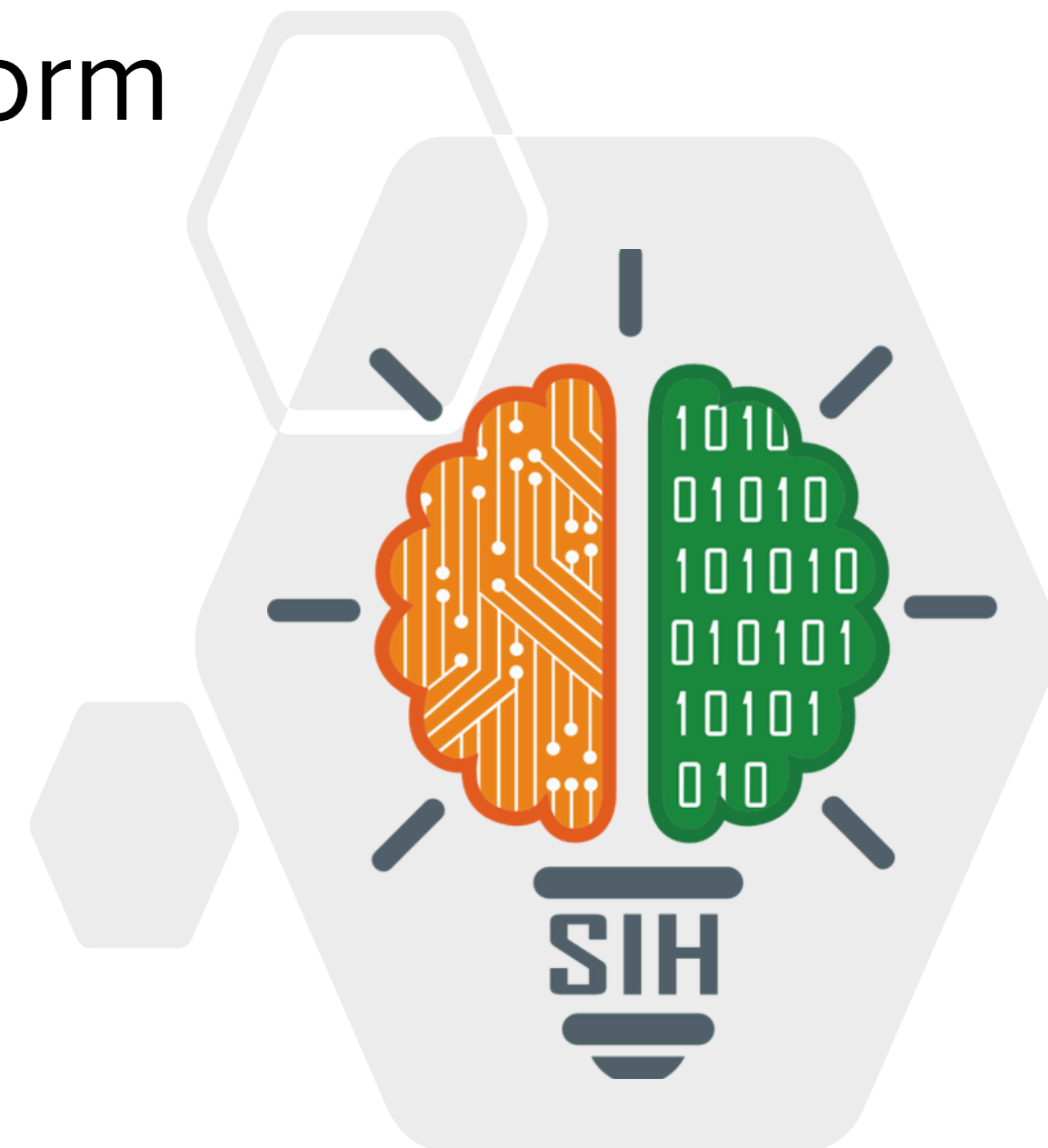
Problem Statement Title - AI Powered Email Threat Detection, Geolocation and Forensic Platform

Theme - Blockchain & Cybersecurity

PS Category - Software

Team ID -

Team Name - STRIKERS



What is the real problem?

India saw a **175% surge in financial phishing attacks with 1,35,000+ attacks** in just 6 months (**CERT-In**). Existing email security tools detect spam but cannot trace the attacker's origin, validate forensic evidence, or support legal proceedings. Organizations are blind to **WHO attacked them, WHERE from, and HOW to prove it in court.**



Proposed solution:

- Layer 1 - Detection:** Chrome Extension & Android App intercept and detonate **suspicious links before users click**, using a stealth headless Chromium sandbox.
- Layer 2 - Investigation:** SOC Web Portal traces the complete attack path **email headers → SMTP relay hops → IP geolocation → WHOIS domain intelligence → interactive attack graph.**
- Layer 3 - Evidence:** Every analysis is cryptographically sealed with **SHA-256 and exported as a Section 63 BSA (2023)** compliant court admissible forensic dossier.

TARGET USERS



Mobile App

Everyday users
(general public)



Chrome Extension

Corporates &
Institutions

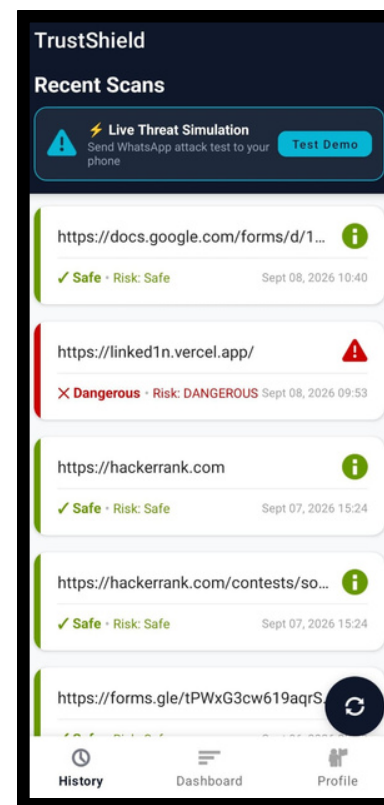


SOC Web Portal

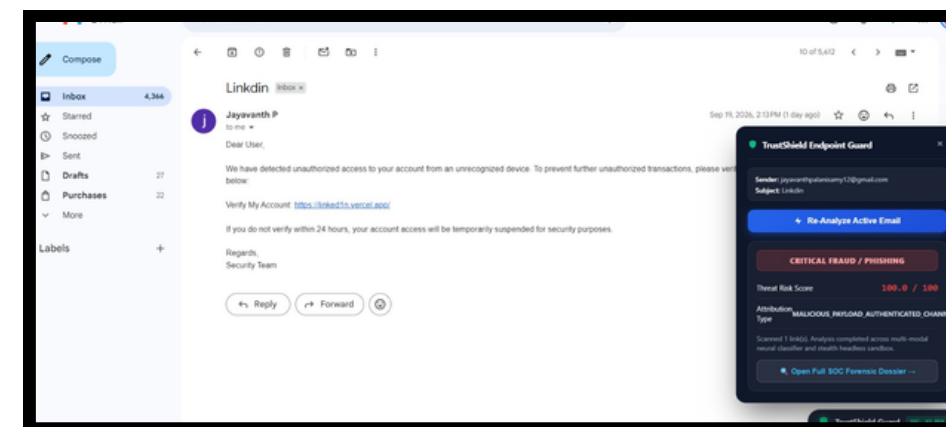
Cyber Investigators &
Law Enforcement

SOC = Security Operations Center - centralized cybersecurity monitoring unit.

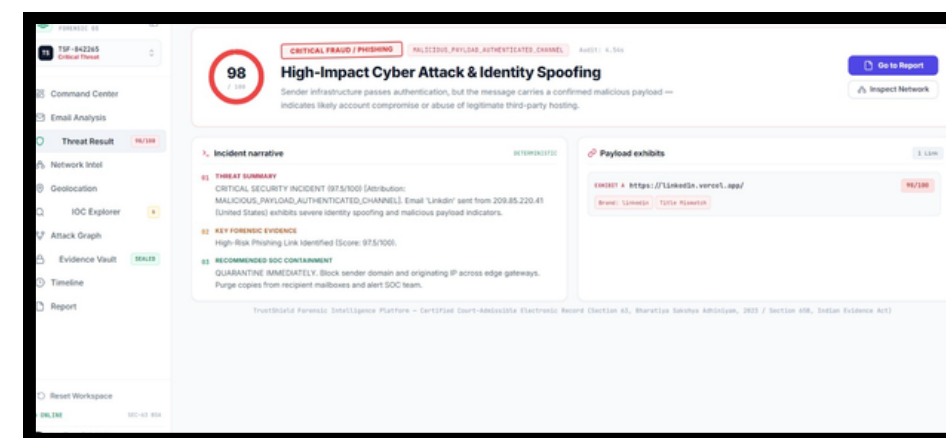
Prototype:



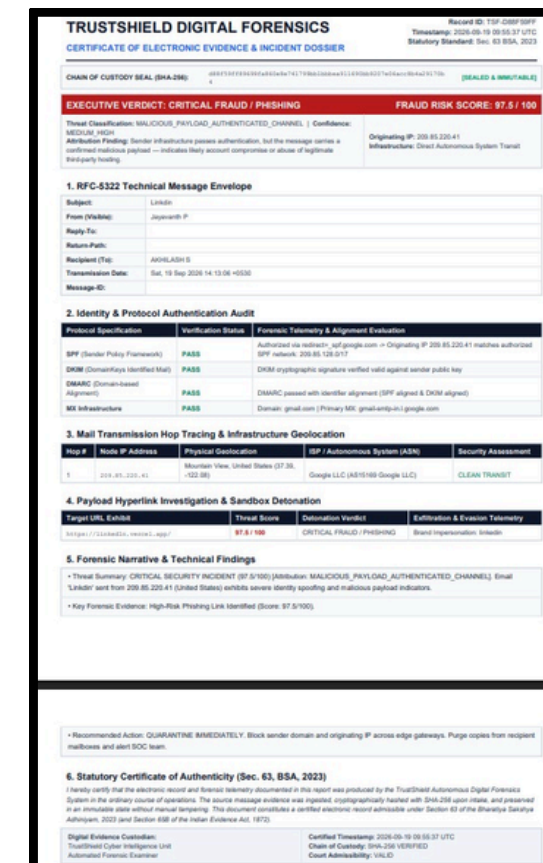
Mobile App



Browser Extension



SOC Investigation Console

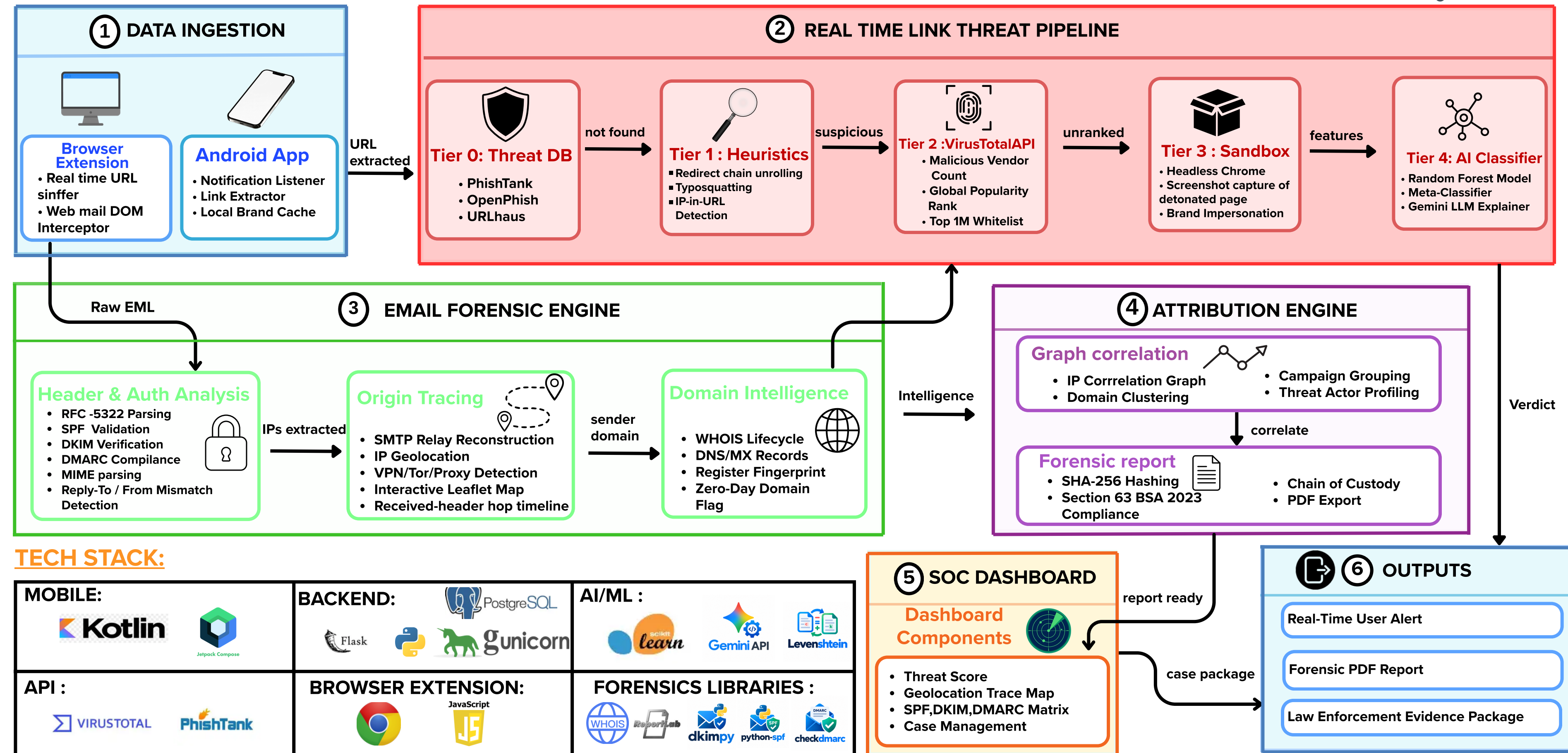


Report – Section 63
(BSA) dossier



Innovation:

- Stealth Sandbox Detonation** - Fires live URLs in isolated Chromium to catch **zero-day credential harvesting pages** no blacklist knows about.
- Attribution, Not Just Detection** - Traces attacker infrastructure across **IP hops, WHOIS data, and relay chains** to expose who is behind the attack.
- One Engine, Three Deployments** - Shared forensic core powers **Android app, Chrome extension, and SOC web portal** simultaneously.
- Court Ready by Design** - SHA-256 tamper evident sealing + **Section 63 BSA (2023)** digital evidence compliance built into every analysis.



TECH STACK:

MOBILE: 	BACKEND: 	AI/ML :
API : 	BROWSER EXTENSION: 	FORENSICS LIBRARIES :

FEASIBILITY AND VIABILITY

Challenges

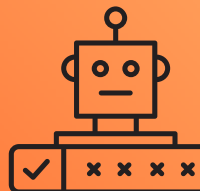


Strategy

Real-time analysis of zero-day phishing links often causes high system latency.



Advanced attackers use anti-bot traps and dynamic domains to evade automated scanning.



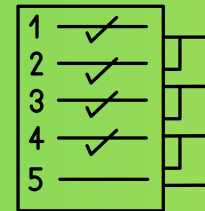
Tracing true attacker origins through spoofed and anonymized SMTP relays is highly complex.



Ensuring automated digital evidence is strictly admissible in a court of law.



Implement a tiered 5 - stage AI pipeline with local heuristics for millisecond execution.



Deploy an evasive headless sandbox that mimics human interaction to bypass traps.



Deep-parse RFC-5322 headers and map IP hops to an interactive IP Geolocation map.



Apply automated SHA - 256 cryptographic hashing to generate Section 63 (BSA) - compliant reports



Feasibility:



Technical Viability – Lightweight client + cloud-deployed sandbox; heavy analysis offloaded to backend via proven OSINT APIs.



Operational Efficiency - 5-stage tiered pipeline resolves 80% of threats locally in under 200ms, minimizing latency and battery drain.



Legal Compliance – SHA-256 hashing protects evidence integrity – Section 63 BSA 2023 compliant court-ready documentation.



Market Need – Helps SOC teams and law enforcement replace manual log analysis with automated threat attribution.

Implementation plan :

Phase 1



Core Detection: Real-time Android link interception 5-tier AI threat classification pipeline Gemini LLM threat explainer

Phase 2



Forensic Portal: SOC dashboard with EML parsing SPF / DMARC / DKIM & relay tracing Geolocation map with VPN flags

Phase 3



Attribution & Deploy: Graph-based campaign attribution Section 63 BSA forensic PDF reports Cloud deployment

WHY NOW?

- India: 17.9% rise in cybercrime, 2024 (NCRB)
- Bank fraud tripled to ₹36,014 Cr in FY25 (RBI)



- New BSA 2023 digital evidence law.
- AI-generated phishing attacks rising.
- UPI & digital payments booming.

Use cases:

LAW ENFORCEMENT & CYBER FORENSICS



- Generates SHA-256 hashed, tamper-proof dossiers compliant with **Section 63 BSA for legal admissibility**.
- Pinpoint the attacker's true origin on an interactive dashboard using **SMTP relay chain reconstruction**.

ENTERPRISE SOC ANALYSTS



- Instant **.eml analysis** to detect BEC, Reply-To spoofing, and DMARC failures.
- Instantly **extract threat indicators (IOCs)** and visualize hidden Tor/VPN proxies from a single dashboard.

FINANCIAL INSTITUTIONS (BANKS)



- Prevent credential theft and brand impersonation by detonating **suspicious links in a stealth sandbox**.
- Block wire fraud and BEC attacks with **zero-click interception** before a transaction occurs.

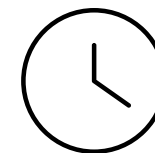
Idea Impact:

- Proactive zero-click interception **prevents financial loss before a single click**, login, or transfer occurs.
- Turns phishing alerts into actionable leads, empowering teams to trace, **report, and dismantle cybercriminal infrastructure** through verified forensic leads.

UNSD Goals:



Benefits:



- Reduced Investigation Time:** Reduces forensic investigation from 3+ hours to **under 60 seconds**.

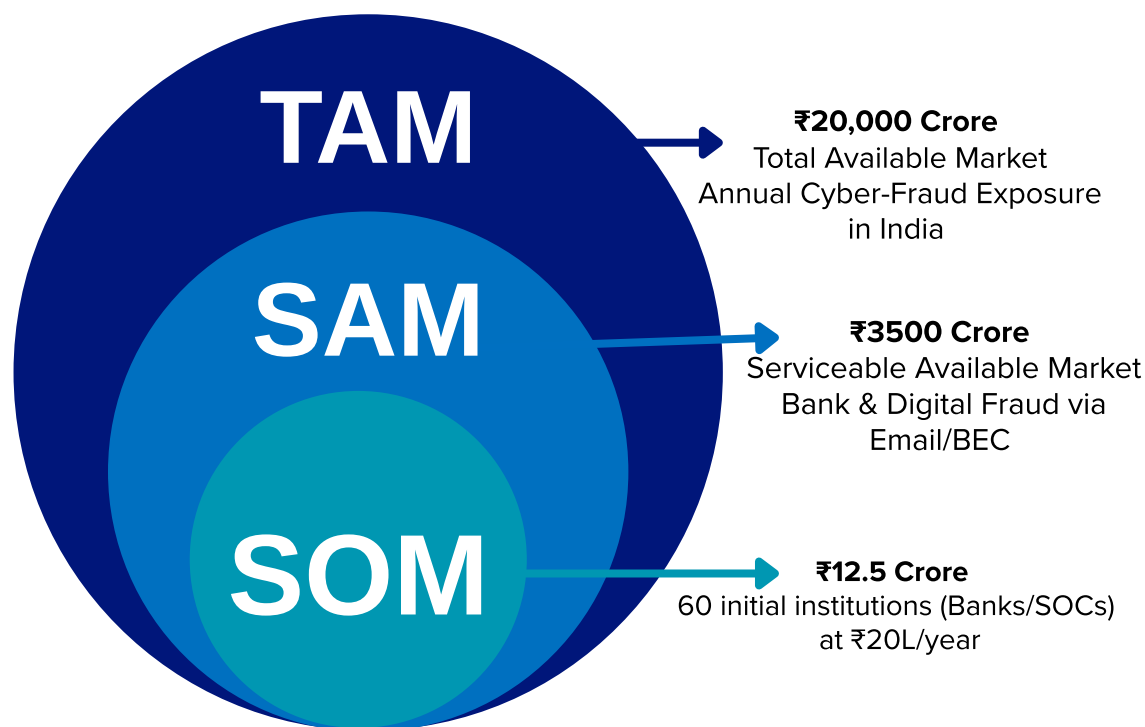


- Lower Fraud Losses:** Proactive sandbox detonation blocks BEC and wire fraud **before any money is lost**.



- Fewer False Positives:** **Dynamic brand verification** easily spots lookalikes, reducing analyst alert fatigue.

Business Model



Progress Report :

Detection engine, Forensic Portal
& BSA dossier - Live | Mobile App
UI & Attribution - In Progress



Sno	Target Customer	Annual Cost
1	Large Banks & Govt	₹12,00,000/-
2	Mid-Size Fintechs	₹5,00,000/-
3	Law Enforcement	₹2,50,000/-
4	Small Businesses	₹1,50,000/-
5	3rd Party Integrations	Pay-per-use

COMPETITIVE ANALYSIS:

Criteria	Standard Gateways (Proofpoint/ Barracuda)	Generic AI Wrappers (ChatGPT / LLMs)	TrustShield
Protocol Audit (SPF/DKIM/DMARC)	✓	✗	✓
Reverse Hop & Tor Tracing	✗	✗	✓
Form Exfiltration Sandbox	✗	✗	✓
Multi-Track Attribution	✗	✗	✓
Legal PDF Dossier (Sec 63 BSA 2023)	✗	✗	✓

REFERENCES:

- ic3.gov/PSA/2024/PSA240911
- <https://csrc.nist.gov/publications/detail/sp/800-177/rev-1/final>
- <https://cert-in.org.in/>
- <https://apwg.org/trendsreports/>
- <https://www.ncrb.gov.in/>
- <https://www.verizon.com/business/resources/reports/dbir/>

Kindly Check Our Additional
Documents & links for the Project

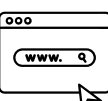
[MORE INFO](#)



- [GitHub Repository](#)



- [Google Drive](#)



- [Web Portal](#)

- Cloud Hosting (Render/AWS): ₹60,000 / Year
- Database (PostgreSQL): ₹15,000 / Year
- 3rd Party APIs (LLM/Threat Feeds): ₹25,000 / Year

Total Expected Expense: ₹1,00,000 / Year